Calcium: Fact Sheet for Consumers (original version)

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What is calcium and what does it do?

Calcium is a mineral your body needs to build and maintain strong bones and to carry out many important functions. Calcium is the most abundant mineral in the body.

Almost all calcium in the body is stored in bones and teeth, giving them structure and hardness.

Your body needs calcium for muscles to move and for nerves to carry messages between your brain and every part of your body. Calcium also helps blood vessels move blood throughout your body and helps release hormones that affect many functions in your body. Vitamin D helps your body absorb calcium.

How much calcium do I need?

The amount of calcium you need each day depends on your age and sex. Average daily recommended amounts are listed below in milligrams (mg).

Life Stage	Recommended Amount
Birth to 6 months	200 mg
Infants 7–12 months	260 mg
Children 1–3 years	700 mg
Children 4–8 years	1,000 mg
Children 9–13 years	1,300 mg
Teens 14–18 years	1,300 mg
Adults 19–50 years	1,000 mg
Adult men 51–70 years	1,000 mg
Adult women 51–70 years	1,200 mg
Adults 71 years and older	1,200 mg
Pregnant and breastfeeding teens	1,300 mg
Pregnant and breastfeeding women	1,000 mg

What foods provide calcium?

Calcium is found in many foods. You can get recommended amounts of calcium by eating a variety of foods, including the following:

- Milk, yogurt, and cheese are the main food sources of calcium for most people in the United States.
- Canned sardines and salmon with bones contain calcium.
- Certain vegetables such as kale, broccoli, and Chinese cabbage (bok choi) also contain calcium.
- Calcium is added to some beverages, including many fruit juices and milk substitutes such as soy and almond beverages, as well as some brands of tofu and ready-to-eat cereals. To find out whether these foods have calcium added, check the product labels.
- Most grains (such as breads, pastas, and unfortified cereals) do not have high amounts of calcium. However, because people eat them often, what they contribute adds up.

What kinds of calcium dietary supplements are available?

Calcium is found in many multivitamin/mineral supplements, in calcium supplements, and in supplements that contain calcium and other nutrients such as vitamin D. Check the Supplement Facts label to determine the amount of calcium in the supplement.

The two main forms of calcium in dietary supplements are calcium carbonate and calcium citrate. Calcium carbonate is absorbed best when taken with food. Some over-the-counter antacids, such as Tums and Rolaids, also contain calcium carbonate.

Calcium citrate is absorbed well on an empty stomach or a full stomach. People with low levels of stomach acid—a condition most common in older people—absorb calcium citrate more easily than calcium carbonate.

Other forms of calcium in supplements and fortified foods include calcium sulfate, calcium ascorbate, calcium microcrystalline hydroxyapatite, calcium gluconate, calcium lactate, and calcium phosphate.

Calcium is absorbed best when you take 500 mg or less at one time. If you take 1,000 mg/day of calcium from supplements, for example, it is better to take a smaller dose twice a day than to take it all at once.

Calcium supplements might cause gas, bloating, and constipation in some people. If you have any of these symptoms, try spreading out the calcium dose throughout the day, taking the supplement with meals, or switching the form of calcium you take.

Am I getting enough calcium?

Many people in the United States get less than recommended amounts of calcium from food and supplements, especially:

- Children and teens age 4 to 18 years
- People who are Black or Asian
- Adults age 50 years and older living in poverty

Certain groups of people are more likely than others to have trouble getting enough calcium, including:

- Postmenopausal women. The body absorbs and retains less calcium after menopause. Over time, this can lead to fragile bones.
- People who don't drink milk or eat other dairy products. Dairy products are rich sources of calcium, but people with lactose intolerance, people with milk allergies, and vegans (people who don't consume any animal products) must find other sources of calcium. Options include lactose-free or reduced-lactose dairy products; canned fish with bones; certain vegetables, such as kale, broccoli, and Chinese cabbage; calcium-fortified fruit juices and milk substitutes such as soy and almond beverages, tofu, and ready-to-eat cereals; and dietary supplements that contain calcium.

What happens if I don't get enough calcium?

Getting too little calcium can cause several conditions, including the following:

- Osteoporosis, which causes weak, fragile bones and increases the risk of falls and fractures (broken bones)
- Rickets, a disease in children that causes soft, weak bones
- Osteomalacia, which causes soft bones in children and adults

What are some effects of calcium on health?

Scientists are studying calcium to understand how it affects health. Here are several examples of what this research has shown.

Bone health in older adults

After about age 30, bones slowly lose calcium. In middle age, bone loss speeds up and can lead to weak, fragile bones (osteoporosis) and broken bones. Although bone loss is more common in women, it can affect men too.

The health of your bones is measured with a bone mineral density test, which will tell whether your bones are healthy and strong or weak and thin. Some studies have found that calcium supplements with or without vitamin D increase bone mineral density in older adults, but others do not. In addition, it is not clear whether calcium supplements help prevent fractures. More research is needed to better understand whether consuming more calcium from food or supplements improves bone health in older adults.

Cancer

Some research shows that people who have high intakes of calcium from food and supplements have a lower risk of cancers of the colon and rectum, but other studies do not. Some studies have shown that men with high intakes of calcium from dairy foods have an increased risk of prostate cancer. For other types of cancer, calcium does not appear to affect the risk of getting cancer or dying of cancer. More research is needed to better understand whether calcium from foods or dietary supplements affects cancer risk.

Heart disease

Calcium can attach to fats and reduce the amount of fat that your body absorbs. Some studies show that calcium supplements have no effect on heart disease, while others show calcium supplements might even increase the risk of heart disease. Overall, experts believe that calcium intakes with or without vitamin D from foods or supplements do not affect the risk of heart disease or of dying from heart disease. (See the section called **Can calcium be harmful?**)

Preeclampsia

Preeclampsia is a serious complication of late pregnancy. Symptoms include high blood pressure and high levels of protein in the urine. Calcium supplements might reduce the risk of preeclampsia in some pregnant women who consume too little calcium. Therefore, many experts recommend calcium supplements during pregnancy for women with low calcium intakes.

Weight management

Research hasn’t clearly shown whether calcium from dairy products or supplements helps you lose weight or prevents weight gain. Some studies show that consuming more calcium helps, but other studies do not. For more information, read our fact sheet on dietary supplements for weight loss.

Metabolic syndrome

Metabolic syndrome is a serious medical condition that increases your risk of heart disease, stroke, and diabetes. You have metabolic syndrome if you have three or more of the following:

- a large waistline
- high blood levels of fat (triglycerides)
- low levels of high-density lipoprotein cholesterol (good cholesterol)
- high blood pressure
- high blood sugar levels

Some research suggests that a higher intake of calcium might help lower the risk of metabolic syndrome in women but not men. More studies are needed.

Can calcium be harmful?

Some research suggests that high calcium intakes might increase the risk of heart disease and prostate cancer.

High levels of calcium in the blood and urine can cause poor muscle tone, poor kidney function, low phosphate levels, constipation, nausea, weight loss, extreme tiredness, frequent need to urinate, abnormal heart rhythms, and a high risk of death from heart disease. However, high levels of calcium in the blood and urine are usually caused by a health condition such as high levels of parathyroid hormone or cancer, not by high calcium intakes.

The daily upper limits for calcium include intakes from all sources—food, beverages, and supplements—and are listed below.

Life Stage	Upper Limit
Birth to 6 months	1,000 mg
Infants 7–12 months	1,500 mg
Children 1–8 years	2,500 mg
Children 9–18 years	3,000 mg
Adults 19–50 years	2,500 mg
Adults 51 years and older	2,000 mg
Pregnant and breastfeeding teens	3,000 mg
Pregnant and breastfeeding women	2,500 mg

Does calcium interact with medications or other dietary supplements?

Calcium dietary supplements can interact or interfere with certain medicines, and some medicines can lower calcium levels in your body. Here are some examples.

- Dolutegravir (Dovato, Tivicay) is a medicine used to treat HIV. Taking calcium supplements at the same time as dolutegravir can lower blood levels of the medicine. To help avoid this interaction, take dolutegravir 2 hours before or 6 hours after taking calcium supplements.
- Levothyroxine (Synthroid, Levoxyl, and others) is a thyroid hormone used to treat hypothyroidism and thyroid cancer. Levothyroxine is not absorbed well when taken within 4 hours of taking a calcium carbonate supplement.
- Lithium (Eskalith, Lithobid) is used to treat bipolar disorder. Long-term use of lithium, or taking lithium together with calcium supplements, can lead to abnormally high levels of calcium in your blood.

- Quinolone antibiotics (examples are ciprofloxacin [Cipro], gemifloxacin [Factive], and moxifloxacin [Avelox]) are not absorbed well when taken within 2 hours of taking a calcium supplement.

Tell your doctor, pharmacist, and other health care providers about any dietary supplements and prescription or over-the-counter medicines you take. They can tell you if the dietary supplements might interact with your medicines or if the medicines might interfere with how your body absorbs, uses, or breaks down nutrients such as calcium.

Calcium and healthful eating

People should get most of their nutrients from food and beverages, according to the federal government's *Dietary Guidelines for Americans*. Foods contain vitamins, minerals, dietary fiber, and other components that benefit health. In some cases, fortified foods and dietary supplements are useful when it is not possible to meet needs for one or more nutrients (for example, during specific life stages such as pregnancy). For more information about building a healthy dietary pattern, see the *Dietary Guidelines for Americans*.

Where can I find out more about calcium?

- For general information on calcium
  - Office of Dietary Supplements (ODS) Health Professional Fact Sheet on Calcium
  - Calcium and Calcium in diet, MedlinePlus
- For more information on food sources of calcium
  - USDA's FoodData Central
  - Nutrient List for calcium (listed by food or by calcium content), USDA
- For more advice on buying dietary supplements
  - ODS Frequently Asked Questions: Which brand(s) of dietary supplements should I purchase?
- For information about building a healthy dietary pattern
  - Dietary Guidelines for Americans

Glossary

absorption
: In nutrition, the process of moving protein, carbohydrates, fats, and other nutrients from the digestive system into the bloodstream. Most absorption occurs in the small intestine.

antibiotic
: A drug used to treat infections caused by bacteria and other microorganisms.

blood sugar
: The main source of energy used by the body's cells. Blood sugar comes from food and is made by the liver, and is carried to the cells through the bloodstream. Also called blood glucose.

blood vessel
: A tube through which blood circulates in the body. Blood vessels include a network of arteries, arterioles, capillaries, venules, and veins.

bone density
: A measurement of bone mass and an indicator of bone strength and health. Also called bone mineral density.

calcium
: A mineral found throughout the body. Calcium is needed for healthy bones and teeth, for nerves and enzymes to function properly, and for blood clotting. Calcium is found in some foods, including milk, yogurt, and cheese, and in Chinese cabbage, kale, broccoli and fortified foods, such as many drinks, tofu, and cereals.

calcium carbonate
: A chemical compound naturally found in chalk, some seashells and other substances. Calcium carbonate is used in antacid drugs to treat indigestion and as a source of calcium to supplement the diet.

cholesterol

A substance found throughout the body. It is made by the liver and is an important component of cells. Cholesterol is also used to make hormones, bile acid, and vitamin D. Foods that come from animals contain cholesterol, including eggs, dairy products, meat, poultry and fish. High blood levels of cholesterol increase a person's chance (risk) of developing atherosclerosis and heart disease.

colon
: A tube-like organ about 5 feet long in adults that is connected to the small intestine at one end and the anus at the other. The colon absorbs water, some nutrients, and electrolytes (such as sodium and chloride) from partially digested food. The remaining material (solid waste called stool) moves through the colon to the rectum and leaves the body through the anus as a bowel movement. The colon is part of the digestive system (a series of organs from the mouth to the anus). Also called the large intestine.

complication
: In medicine, an illness or condition that occurs while a patient has a disease. The complication is not a part of the disease, but may be a result of the disease or may be unrelated.

constipation
: A condition in which stool becomes hard, dry, and difficult to pass and bowel movements happen infrequently. Other symptoms may include painful bowel movements and feeling bloated, uncomfortable, and sluggish.

consume
: To eat or drink.

dairy food
: Milk and products made with milk, such as buttermilk, yogurt, cheese, cottage cheese, and ice cream.

diabetes
: A disease in which blood sugar (glucose) levels are high because the body is unable to use glucose properly. Diabetes occurs when the body does not make enough insulin, which helps the cells use glucose, or when the body no longer responds to insulin.

dietary fiber
: A substance in plants that you cannot digest. It adds bulk to your diet to make you feel full, helps prevent constipation, and may help lower the risk of heart disease and diabetes. Good sources of dietary fiber include whole grains (such as brown rice, oats, quinoa, bulgur, and popcorn), legumes (such as black beans, garbanzo beans, split peas, and lentils), nuts, seeds, fruit, and vegetables.

Dietary Guidelines for Americans
: Advice from the federal government to promote health and reduce the chance (risk) of long-lasting (chronic) diseases through nutrition and physical activity. The Guidelines are updated and published every 5 years by the US Department of Health and Human Services and the US Department of Agriculture.

dietary supplement
: A product that is intended to supplement the diet. A dietary supplement contains one or more dietary ingredients (including vitamins, minerals, herbs or other botanicals, amino acids, and other substances) or their components; is intended to be taken by mouth as a pill, capsule, tablet, or liquid; and is identified on the front label of the product as being a dietary supplement.

disorder
: In medicine, a disturbance of normal functioning of the mind or body. Disorders may be caused by genetic factors, disease, or trauma.

dose
: The amount of medicine or other substance taken at one time or over a specific period of time.

fortified
: When nutrients (such as vitamins and minerals) are added to a food product. For example, when calcium is added to orange juice, the orange juice is said to be "fortified with calcium". Similarly, many breakfast cereals are "fortified" with several vitamins and minerals.

fracture
: A break, for example, a bone fracture.

fragile
: Easily broken.

heart rhythm
: The regular beating of the heart as it moves blood throughout the body.

high blood pressure
: A blood pressure measurement of 140/90 mmHg (millimeters of mercury) or higher is considered high blood pressure (hypertension). Blood pressure is the force of blood pushing against the walls of the

arteries. Blood pressure measurements are written as two numbers, for example 120/80. The first number (the systolic pressure) measures the pressure when the heart beats and pumps out blood into the arteries. The second number (the diastolic pressure) measures the pressure when the heart is at rest between beats. High blood pressure is a condition that occurs when a person's blood pressure often measures above 140/90 or regularly stays at that level or higher. This condition usually has no symptoms but can be life-threatening. It damages the arteries and increases the chance of stroke, heart attack, kidney failure, and blindness. Also called hypertension.

hormone
: A group of chemicals made by glands in the body. Hormones circulate in the bloodstream and control the actions of certain cells or organs. Some hormones can also be manufactured.

hypothyroidism
: A disorder in which the thyroid gland makes too little hormone for the body to function well. Thyroid hormones affect chemical reactions in the body, brain development, breathing, heart and nervous system functions, body temperature, muscle strength, skin dryness, menstrual cycles, cholesterol levels, and body weight.

infant
: A child younger than 12 months old.

interaction
: A change in the way a dietary supplement acts in the body when taken with certain other supplements, medicines, or foods, or when taken with certain medical conditions. Interactions may cause the dietary supplement to be more or less effective, or cause effects on the body that are not expected.

kidney
: One of two organs that remove waste from the blood (as urine). The kidneys also make erythropoietin (a substance that stimulates red blood cell production) and help regulate blood pressure. The kidneys are located near the back under the lower ribs.

label
: When referring to dietary supplements, information that appears on the product container, including a descriptive name of the product stating that it is a "supplement"; the name and place of business of the manufacturer, packer, or distributor; a complete list of ingredients; and each dietary ingredient contained in the product. Supplements must also include directions for use, nutrition labeling in the form of a Supplement Facts panel that identifies each dietary ingredient contained in the product and the serving size, amount, and active ingredients.

lactose
: A type of sugar found in milk and milk products.

menopause
: The time of life when a woman's menstrual periods stop. A woman is in menopause when she hasn't had a period for 12 months in a row. Also called "change of life."

metabolic
: Having to do with metabolism (all chemical changes that take place in a cell or organism to produce energy and basic materials needed for important life processes).

milligram
: mg. A measure of weight. It is a metric unit of mass equal to 0.001 gram (it weighs 28,000 times less than an ounce).

mineral
: In nutrition, an inorganic substance found in the earth that is required to maintain health.

nausea
: The uneasy feeling of having an urge to throw up (vomit).

nerve
: A bundle of microscopic fibers that carries messages back and forth from the brain to other parts of the body.

nutrient
: A chemical compound in food that is used by the body to function and maintain health. Examples of nutrients include proteins, fats, carbohydrates, vitamins, and minerals.

Office of Dietary Supplements
: ODS, Office of Disease Prevention, Office of Director, National Institutes of Health, Department of Health and Human Services. ODS strengthens knowledge and understanding of dietary supplements by evaluating scientific information, stimulating and supporting research, disseminating research results, and educating the public to foster an enhanced quality of life and health for the US population.

osteomalacia
: A condition in adults in which bones become soft and deformed because they don’t have enough calcium and phosphorus. It is usually caused by not having enough vitamin D in the diet, not getting enough sunlight, or a problem with the way the body uses vitamin D. Symptoms include bone pain and muscle weakness. When the condition occurs in children, it is called rickets.

osteoporosis
: A condition in which bones become weak and brittle, increasing the chance they may break.

pharmacist
: A person licensed to make and dispense (give out) prescription drugs and who has been taught how they work, how to use them, and their side effects.

postmenopausal
: Having to do with the time after menopause. The time in a woman's life when menstrual periods stop permanently is called menopause ("change of life").

prescription
: A written order from a health care provider for medicine, therapy, or tests.

prevent
: To stop from happening.

prostate cancer
: Uncontrolled growth of abnormal cells in the prostate (a gland in the male reproductive system found below the bladder and in front of the rectum). Prostate cancer usually occurs in older men.

protein
: A molecule made up of amino acids that the body needs for good health. Proteins are the basis of body structures such as skin and muscle, and substances such as enzymes and antibodies.

rickets
: A condition in children in which bones become soft and deformed because they don’t have enough calcium and phosphorus. It is caused by not having enough vitamin D in the diet or by not getting enough sunlight. In adults, this condition is called osteomalacia.

risk
: The chance or probability that a harmful event will occur. In health, for example, the chance that someone will develop a disease or condition.

soy
: A plant that produces beans used in many food products. Soy products contain isoflavones (estrogen-like substances) that are being studied in the prevention of cancer, hot flashes that occur with menopause, and osteoporosis (loss of bone density). Also called soya and soybean. Latin name: *Glycine max*.

stroke
: A loss of blood flow to part of the brain. Strokes are caused by blood clots or broken blood vessels in the brain, and result in damage to a section of brain tissue. Symptoms include dizziness, numbness, weakness on one side of the body, and problems with talking or understanding language. The chance (risk) of stroke is increased by high blood pressure, older age, smoking, diabetes, high cholesterol, heart disease, a family history of stroke, and a build-up of fatty material inside the coronary arteries (atherosclerosis). See also NIH publication: Know Stroke. Know the Signs. Act in Time. http://www.ninds.nih.gov/disorders/stroke/knowstroke.htm

supplement
: A nutrient that may be added to the diet to increase the intake of that nutrient. Sometimes used to mean dietary supplement.

symptom
: A feeling of sickness that an individual can sense, but that cannot be measured by a healthcare professional. Examples include headache, tiredness, stomach ache, depression, and pain.

thyroid
: A gland located in the front of the neck, below the larynx (Adam’s apple). The thyroid makes hormones that circulate in the bloodstream and affect brain development, metabolism, weight, breathing, heart rate, blood pressure, nervous system functions, body temperature, muscle strength, skin dryness, menstrual cycles, and cholesterol levels.

treat
: To care for a patient with a disease by using medicine, surgery, or other approaches.

triglyceride

A type of fat found in your blood. When you eat more than you need, your body turns the excess calories into triglycerides. High blood levels of triglycerides can increase your risk of heart disease, heart attack, and stroke.

upper limit
: UL. The largest daily intake of a nutrient considered safe for most people. Taking more than the UL is not recommended and may be harmful. The UL for each nutrient is set by the Food and Nutrition Board at the National Academies of Sciences, Engineering, and Medicine. For example, the UL for vitamin A is 3,000 micrograms/day. Women who consume more than this amount every day shortly before or during pregnancy have an increased chance (risk) of having a baby with a birth defect. Also called the tolerable upper intake level.

urine
: Excess liquids and wastes that have been filtered from the blood by the kidneys, stored in the bladder, and removed from the body through the urethra (the tube that carries urine from the bladder to outside the body).

US Department of Agriculture
: USDA promotes America's health through food and nutrition, and advances the science of nutrition by monitoring food and nutrient consumption and updating nutrient requirements and food composition data. USDA is responsible for food safety, improving nutrition and health by providing food assistance and nutrition education, expanding markets for agricultural products, managing and protecting US public and private lands, and providing financial programs to improve the economy and quality of rural American life.

vegan
: A person who eats only plant-based foods. Vegans do not eat meat, poultry, fish, eggs, milk, dairy products, or honey, and do not use leather, silk, or wool, or soaps and cosmetics that are made from animal products.

vitamin
: A nutrient that the body needs in small amounts to function and maintain health. Examples are vitamins A, C, and E.

vitamin D
: A nutrient that is obtained from the diet and can be made in the skin after exposure to sunlight. Vitamin D acts as a hormone. It helps to form and maintain strong bones, maintain normal blood levels of calcium and phosphorus, and increase calcium absorption; it also helps to maintain a healthy immune system and control cell growth. Vitamin D is found in some foods, including some types of fatty fish, and milk and breakfast cereals that are fortified with vitamin D.

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